

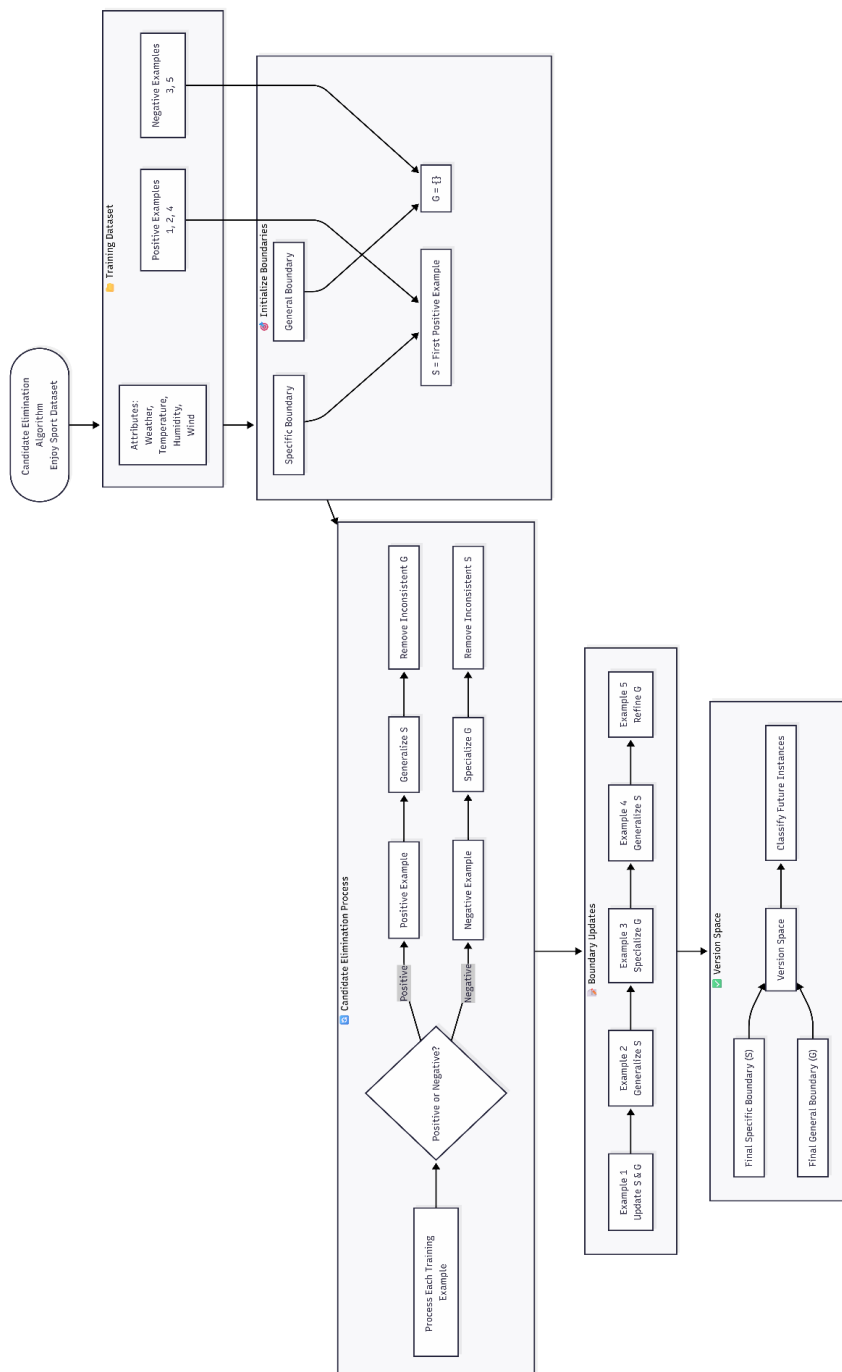
ITA0602 – Machine Learning

CO1 – AT2 – CONCEPT LEARNING

Question 1)

Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities.

- Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).
- Explain feature scaling and handling missing data.
- Describe how overfitting can be detected and avoided.



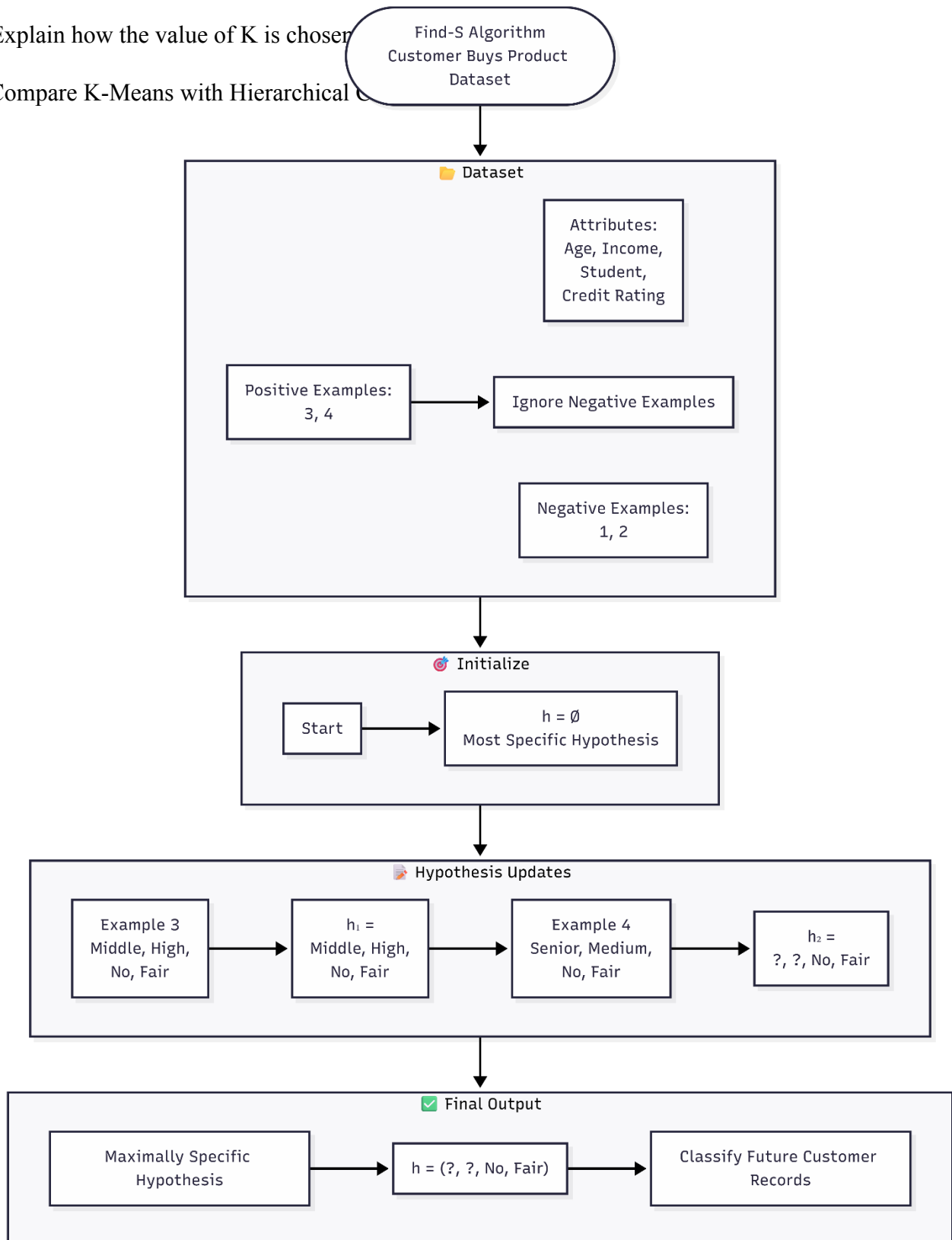
Question 2)

Explain the concept of clustering in unsupervised learning. (5 Marks)

(a) Describe the working of the K-Means algorithm with a step-by-step example.

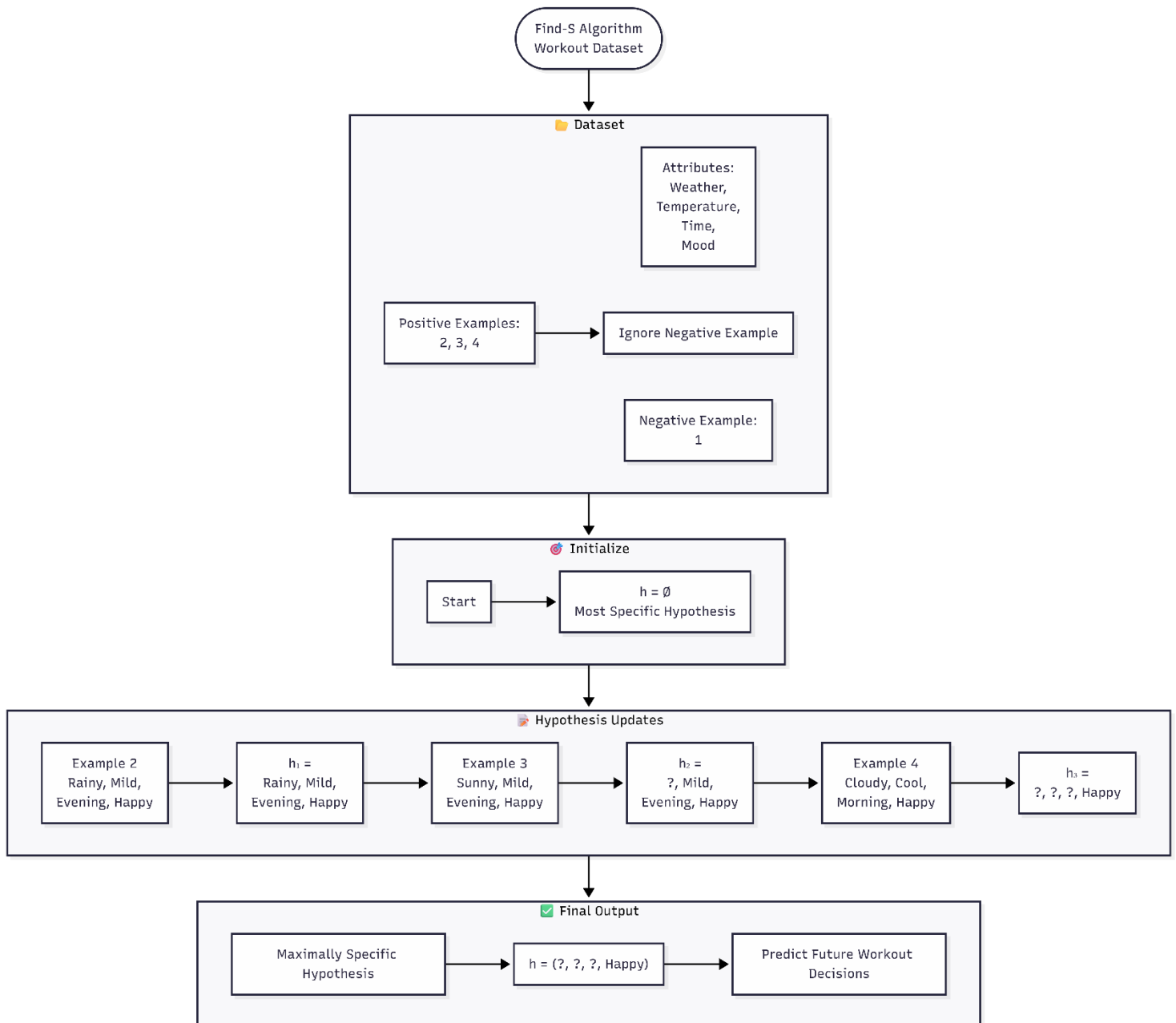
(b) Explain how the value of K is chosen

(c) Compare K-Means with Hierarchical Clustering



Question 3)

With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme

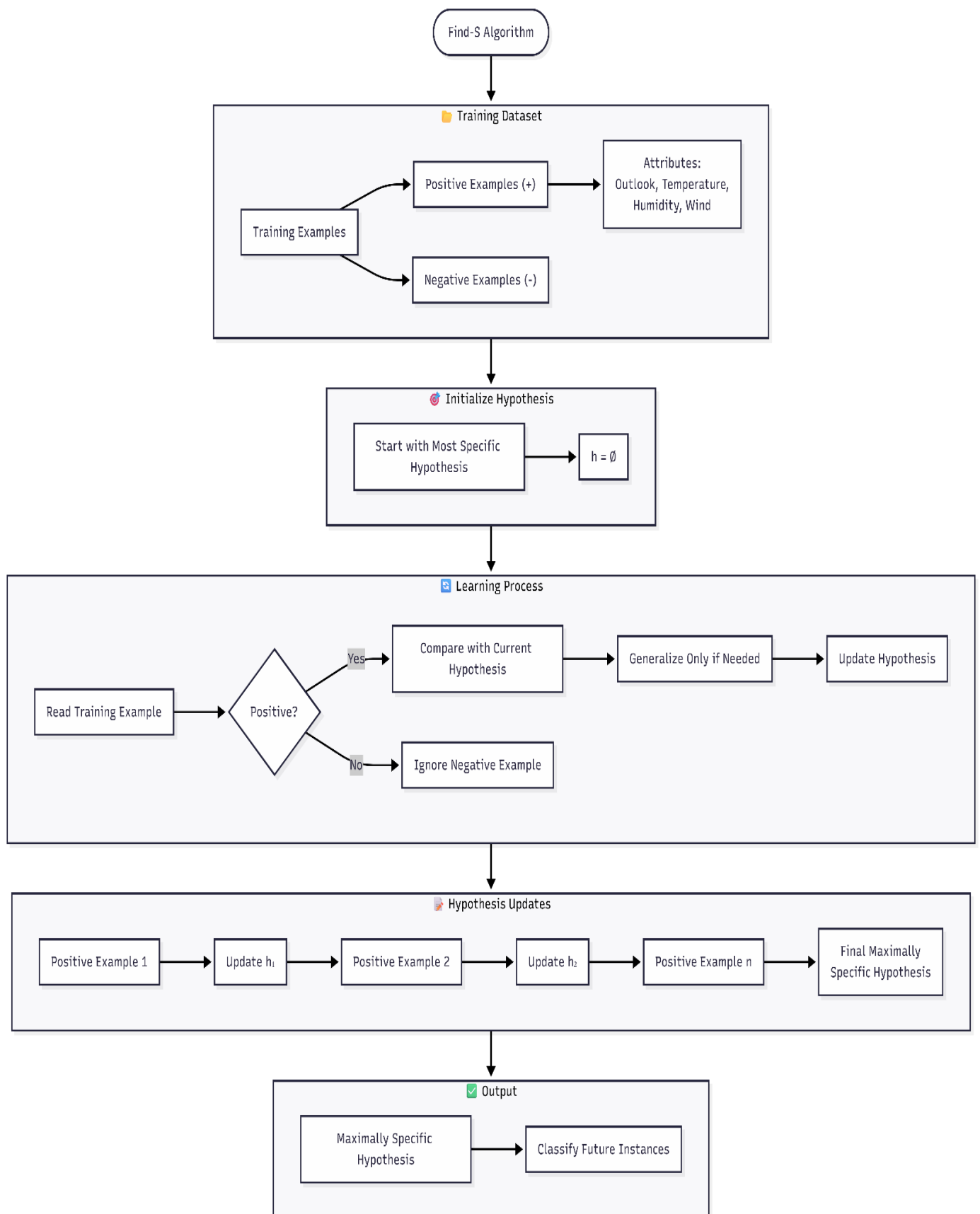


Question 4)

In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating.

Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples.

Finally, provide the final hypothesis.



Question 5)

Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques. (a) Describe the steps involved in training, validating, and evaluating the model. (b) Explain how the system can assist in early warning and emergency response. create concept map document file for this and add images for each questions belongs to the question it could be 15 pages minimum generate a ppt file

